

## Introduction

Healthcare associated infections are a major issue in the quality and safety of health care, in both hospital and community settings. Identification of patients with specific multi drug resistant organisms (MDRO's), can reduce the risk of transmission of these organisms within healthcare and thus reduce morbidity and mortality associated with these organisms.

### 1. Purpose:

This policy provides instruction to staff on indication & sampling procedure for MDRO's and candida auris screening. The current MDRO of concerns in our hospital are the following:

MDRA: Multi-Drug Resistant *Acinetobacter Baumannii*

CRE: Carbapenem Resistant *Enterobacterales*

MRSA: Methicillin Resistant *Staphylococcus aureus*

VRE: Vancomycin Resistant *Enterococci*

MDR Pseudomonas: Multidrug Resistant *Pseudomonas*

ESBL: Extended Spectrum beta Lactamase

The principle objectives of screening for MDRO and candida auris are:

1. To identify patients who are carriers of MDRO and candida auris.
2. To reduce the risk of patients colonized with MDRO and candida auris developing infection
3. To reduce the risk of transmission of MDRO and candida auris to other patients.

## **2. Scope:**

This policy applies to all patients admitted to Royal hospital. The policy should be applied by all staff working at royal hospital.

## **3. Role and responsibilities:**

### **➤ The Infection prevention & Control (IP&C) committee:**

The committee is responsible to oversee the ongoing development of this policy

### **➤ Head of IP&C department:**

Head of IP&C is responsible for overseeing the implementation of the policy, for reporting any concerns and performance to the administration and/or to infection prevention & control committee.

### **➤ Infection Preventionist (IP's):**

IP's are responsible for training, supporting implementation of the policy and monitoring compliance with the policy.

### **➤ Microbiology laboratory:**

The microbiology lab is responsible for releasing the results of all MDRO screening samples. In case of suspicion of an outbreak the head of IP&C must be notified by the lab.

### **➤ Nursing staff:**

Nurses are responsible for ensuring adherence of all the staff to this policy, collecting samples as outlined in this policy and initiate the isolation/precautions as outlined in the isolation policy for suspected and confirmed cases.

### **➤ Medical staff:**

Doctors are responsible to inform their patients in case the screening come positive and educate them.

#### 4. Definitions

➤ **MDRO:** a bacterial strain that is resistant to one or more of antimicrobial agents.

➤ **MDRAB:**

Any *Acinetobacter spp.* testing non-susceptible (i.e., resistant or intermediate) to at least one agent in at least 3 antimicrobial classes of the following 6 classes:

1. Ampicillin/sulbactam
2. Cephalosporins (cefepime, ceftazidime)
3.  $\beta$ -lactam/ $\beta$ -lactam  $\beta$ -lactamase inhibitor combination (piperacillin, piperacillin/tazobactam)
4. Carbapenems (imipenem, meropenem, doripenem)
5. Fluoroquinolones (ciprofloxacin or levofloxacin)
6. Aminoglycosides (gentamicin, tobramycin, or amikacin).

In A'Shifa the alert appears for positive cases MDRA

➤ **CRE:** Any Enterobacterales e.g. *Escherichia coli*, *Klebsiella oxytoca*, *Klebsiella pneumoniae*, or *Enterobacter spp.* testing resistant to imipenem, meropenem, doripenem, or ertapenem by standard susceptibility testing methods (i.e., minimum inhibitory concentrations of  $\geq 4$  mcg/mL for doripenem, imipenem and meropenem or  $\geq 2$  mcg/mL for ertapenem) OR by production of a carbapenemase (i.e., KPC, NDM, VIM, IMP, OXA-48) demonstrated using a recognized test (e.g., polymerase chain reaction, metallo- $\beta$ -lactamase test, modified-Hodge test, Carba-NP). In A'Shifa the alert appears for positive cases **CRE**

- **MRSA:** Includes *S. aureus* cultured from any specimen that tests oxacillin-resistant, cefoxitin-resistant, or methicillin-resistant by standard susceptibility testing methods, or by a laboratory test that is FDA-approved for MRSA detection from isolated colonies; these methods may also include a positive result by any FDA-approved test for MRSA detection from specific sources. In A'Shifa the alert appears for positive cases **MRSA**
- **VRE:** *Enterococcus faecalis*, *Enterococcus faecium*, or *Enterococcus species unspecified* (only those not identified to the species level) that is resistant to vancomycin, by standard susceptibility testing methods or by results from any FDA-approved test for VRE detection from specific specimen sources. In A'Shifa the alert appears for positive cases **VRE**
- **MDR-Pseudomonas:** Any *Pseudomonas* spp. testing non-susceptible (i.e., resistant or intermediate) to at least one agent in at least 3 antimicrobial classes of the following 5 classes:
  - Cephalosporins (cefepime, ceftazidime)
  - $\beta$ -lactam/ $\beta$ -lactam  $\beta$ -lactamase inhibitor combination (piperacillin, piperacillin/tazobactam)
  - Carbapenems (imipenem, meropenem, doripenem)
  - Fluoroquinolones (ciprofloxacin or levofloxacin)
  - Aminoglycosides (gentamicin, tobramycin, or amikacin)

In A'Shifa the alert appears for positive cases **PAER**

➤ **ESBL:**

Are enzymes that mediate resistance to extended-spectrum (third generation) cephalosporins (e.g., ceftazidime, cefotaxime, and ceftriaxone) and monobactams (e.g., aztreonam) but do not affect cephamycins (e.g., cefoxitin and cefotetan) or carbapenems (e.g., meropenem or imipenem).

In A'Shifa the alert appears for positive cases **ESBL**.

- **Candida Auris:** an emerging fungus that is resistant to multiple classes of antifungal medication (eg: azoles, amphotericin B, echonocandins).
- **MDROs colonization risk factors:** presence of tracheostomy, urinary catheter, chronic wound
- **Colonization** means the MDRO are present in or on the body but are not causing illness. Healthy people may carry the bacteria causing MDRO without becoming ill.
- **Infection** means the MDRO are present in or on the body and are causing illness. Symptoms of MDRO may vary depending on the part of the body that is infected. Infection can occur in any part of the body. Infections may affect the skin, lungs, urinary tract or bloodstream

#### 5. MDROs included in the active surveillance

| Active Surveillance                     | Rescreen from the same site if positive from clinical samples |
|-----------------------------------------|---------------------------------------------------------------|
| MRSA, CRE, MDRAB, VRE and Candida auris | ESBL, MDR Pseudomonas                                         |

The active surveillance means screening actively to detect carriers. For MDRO that are not included in the active surveillance if detected in clinical samples the patient is screened from the same site for the purpose of discontinuing isolation only.

#### PAER screening:

There is no active surveillance screening for Multidrug resistant Pseudomonas aeruginosa (PEAR). Rescreening is required for de-isolation purpose only. When the sample is collected for PEAR screening, it is important to specify this in the remark section. Contact Infection prevention & control staff for further advice.

## **ESBL screening:**

There is active screening for ESBL in the NICU only. Patients colonized or infected with ESBL are isolated in the NICU and surgical areas. ESBL rescreening is required for de-isolation of patients at these areas. When the sample is collected for ESBL screening, it is important to specify this in the remark section. Contact Infection prevention & control staff for further advice

## **6. Screening purposes:**

### **7.1 Screening patients at the time of admission:**

For patients to be transferred directly to Royal Hospital the following should be followed:

- If the case is urgent it can be accepted if his/her screening results are not known and the screening is initiated at Royal Hospital. The patient should be under contact precautions till the results of screening tests released. If results are positive, continue contact precautions as per isolation policy
- For non-urgent transfer, enquiring about the screening results is a must before accepting the patient, accordingly the bed can be arranged as for isolation or not, the screening samples has to be taken within 48 hours before transferring.
- If the screening results are not available within 48hours before transfer, screening to be repeated.
- If results from referring hospital is negative standard precautions should be applied. If results are positive, ask the referring hospital to attach the result in the referring notes.

The following table (1) lists the indications and samples for screening patients based on risk factors.

| Type of organism     | Indications for screening                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Screening request and samples                                                                                                                                |
|----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>MDRA/<br/>CRE</b> | <ol style="list-style-type: none"> <li>1. Admission abroad within the last 6 months</li> <li>2. Admission to other local hospital within the last 3 months</li> <li>3. Readmission to RH with risk factors e.g. wounds, tracheostomy, catheter, central line</li> <li>4. Admission to critical unit (AICU, PICU, HD)</li> <li>5. Admission to post cardiac surgery unit after more 5 days stay in other wards</li> </ol>                                                                                          | <p><b><u>CRE/MDRA screening culture</u></b></p> <p>Rectal swab for body screening</p> <p>Urine, swab or respiratory sample according to the risk factors</p> |
| <b>VRE</b>           | <ol style="list-style-type: none"> <li>1. Admission abroad within the last 6 months</li> <li>2. Admission to other local hospital within the last 3 months</li> <li>3. Readmission to RH with risk factors e.g. wounds, tracheostomy, catheter, central line</li> <li>4. Admission to critical unit (AICU, PICU, HD)</li> <li>5. Admission to post cardiac surgery unit after more 5 days stay in other wards</li> <li>6. Monthly screening for chronic oncology, hematology and hemodialysis patients</li> </ol> | <p><b><u>VRE screening culture</u></b></p> <p>Rectal swab for body screening</p> <p>Urine, swab or respiratory sample according to the risk factors</p>      |

|                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                               |
|----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>MRSA</b>          | <ol style="list-style-type: none"> <li>1. Admission abroad within the last 6 months</li> <li>2. Admission to other local hospital within the last 3 months</li> <li>3. Readmission to RH with risk factors e.g. wounds, tracheostomy, catheter, central line</li> <li>4. Admission to critical unit (AICU, PICU, HD)</li> <li>5. Admission to post cardiac surgery unit after more 5 days stay in other wards</li> <li>6. Prior to cardiac surgeries and transplant surgeries (renal and liver).</li> <li>7. Prior to insertion of dialysis line (hemodialysis or peritoneal dialysis)</li> </ol> | <p><b><u>MRSA screening culture</u></b></p> <p>Nasal, axilla and groin swabs for body screening</p> <p>Urine, swab or respiratory sample according to the risk factors</p> <p><b><u>MRSA PCR (Prior to urgent cardiac surgery)</u></b></p> <p>Nasal PCR swab for MRSA PCR</p> |
| <b>Candida Auris</b> | <ol style="list-style-type: none"> <li>1. Admission abroad within the last 6 months</li> <li>2. Adult patient with recent admission to other local hospital within the last 3 months</li> <li>3. Pediatric patients with recent admission to other local hospital endemic with Candida auris within the last 3 months</li> <li>4. Admission to adult critical units</li> </ol>                                                                                                                                                                                                                    | <p><b><u>Canidia auris screening</u></b></p> <p>Axilla and groin swabs for body screening</p> <p>Urine, swab or respiratory sample according to the risk factors</p>                                                                                                          |



## **7.2 Screening patient admitted to NICU:**

The patients admitted to the NICU are screened for CRE, MDRA, MRSA, ESBL Klebsiella and Serratia if meeting the following criteria.

- Baby or mother was admitted to other hospital in the last 3 months
- Unknown/ abandoned baby
- If mother tests positive for an MDRO, screen the baby for the same MDRO.

The screening samples for CRE, MDRA and MRSA are as described above.

The screening samples for ESBL/Serratia are:

- Rectal swab for body screening
- Urine, swab or respiratory sample according to the risk factors

## **7.3 Screening patient for the purpose of deisolation:**

MDROs screening for the purpose of deisolation has to be done while patient is not on antimicrobials. Follow the following algorithms for the purpose of de-isolation:

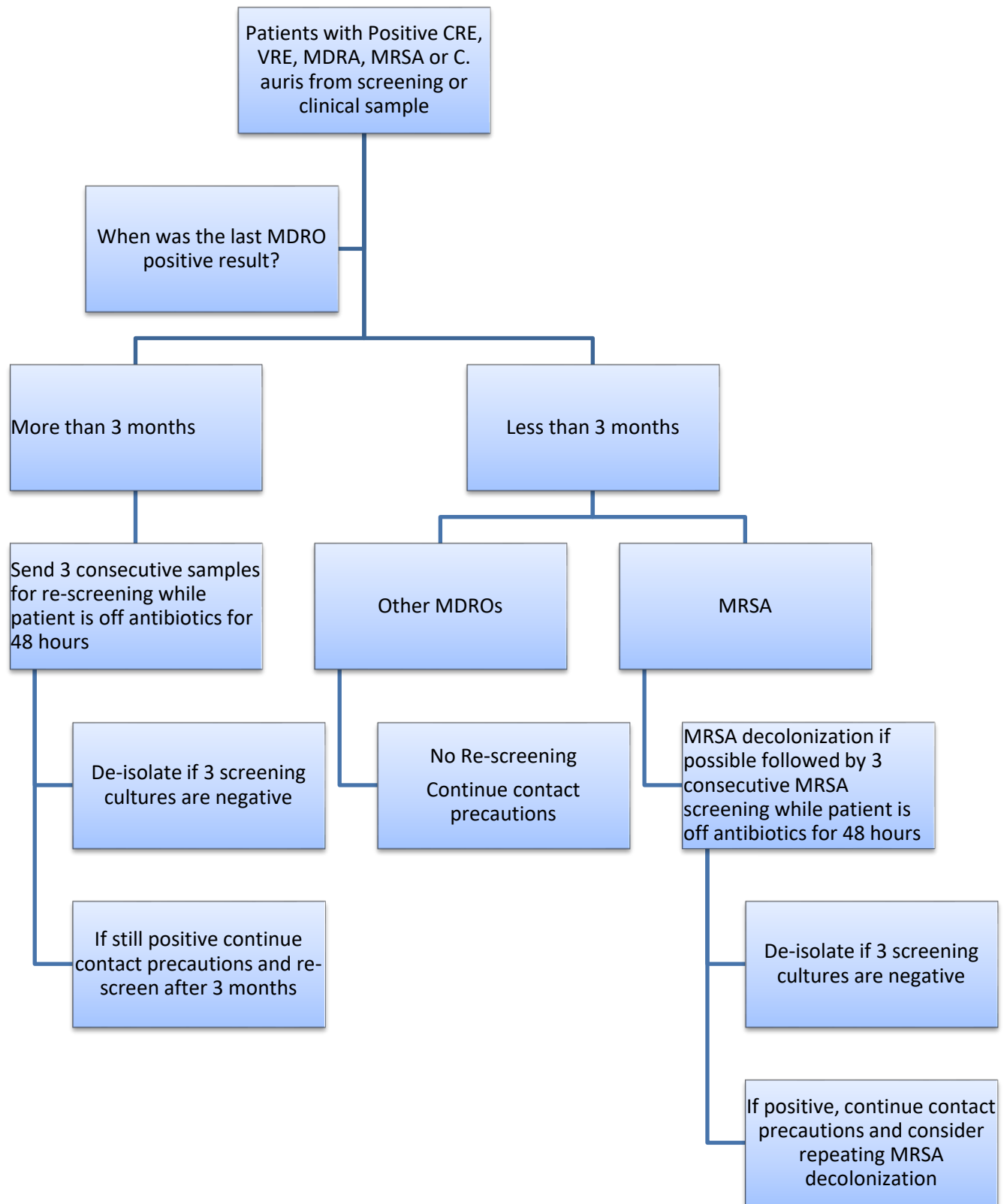


Figure 1: De-isolation of CRE, VRE, MDRA, MRSA and *Candida auris*

If the patient had an ESBL in the surgical areas or MDR *Pseudomonas aeruginosa* in any area, apply the same isolation procedures for other MDROs. Follow the following algorithms for de-isolation:

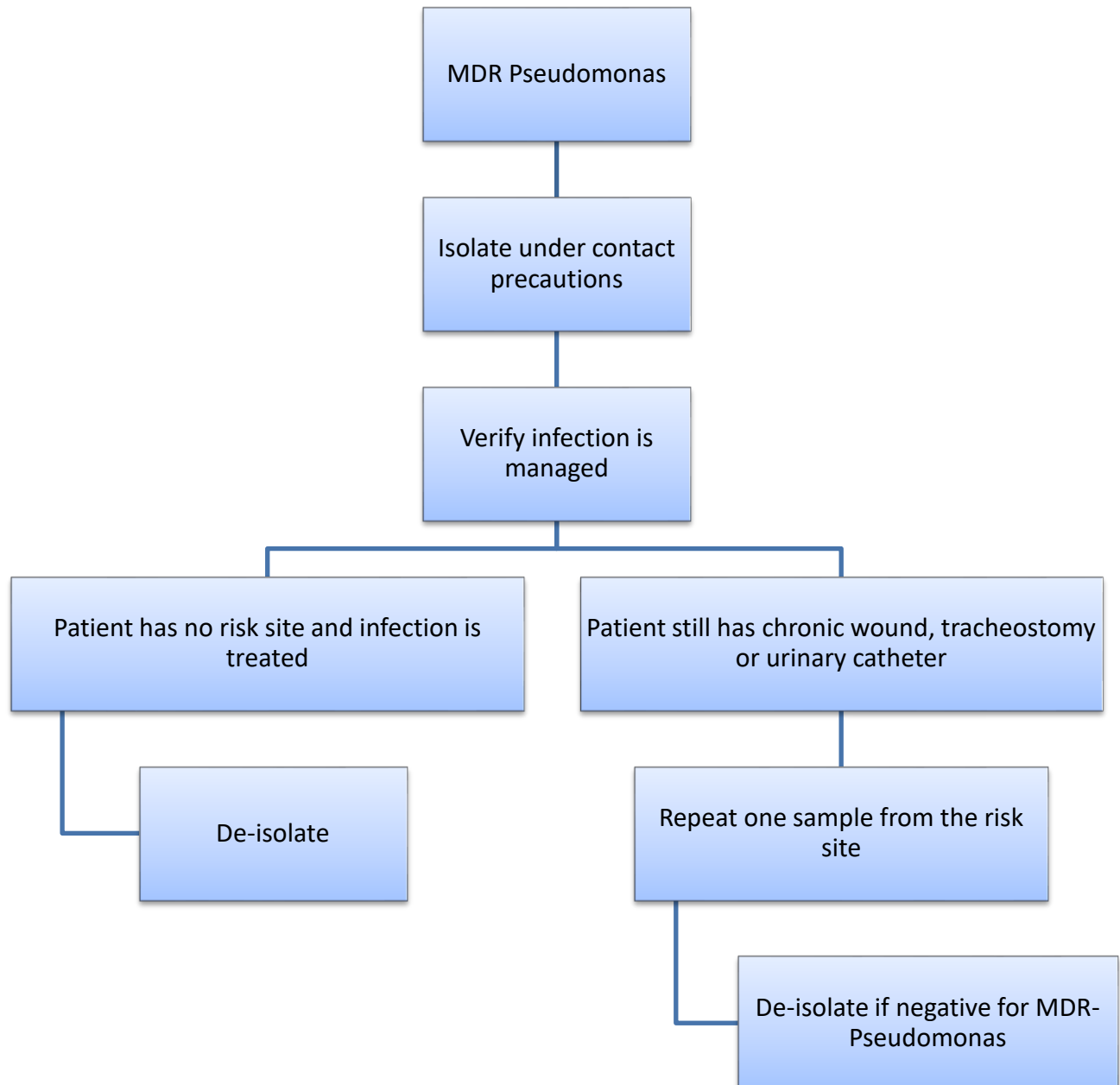


Figure 2: De-isolation of MDR *Pseudomonas aeruginosa*

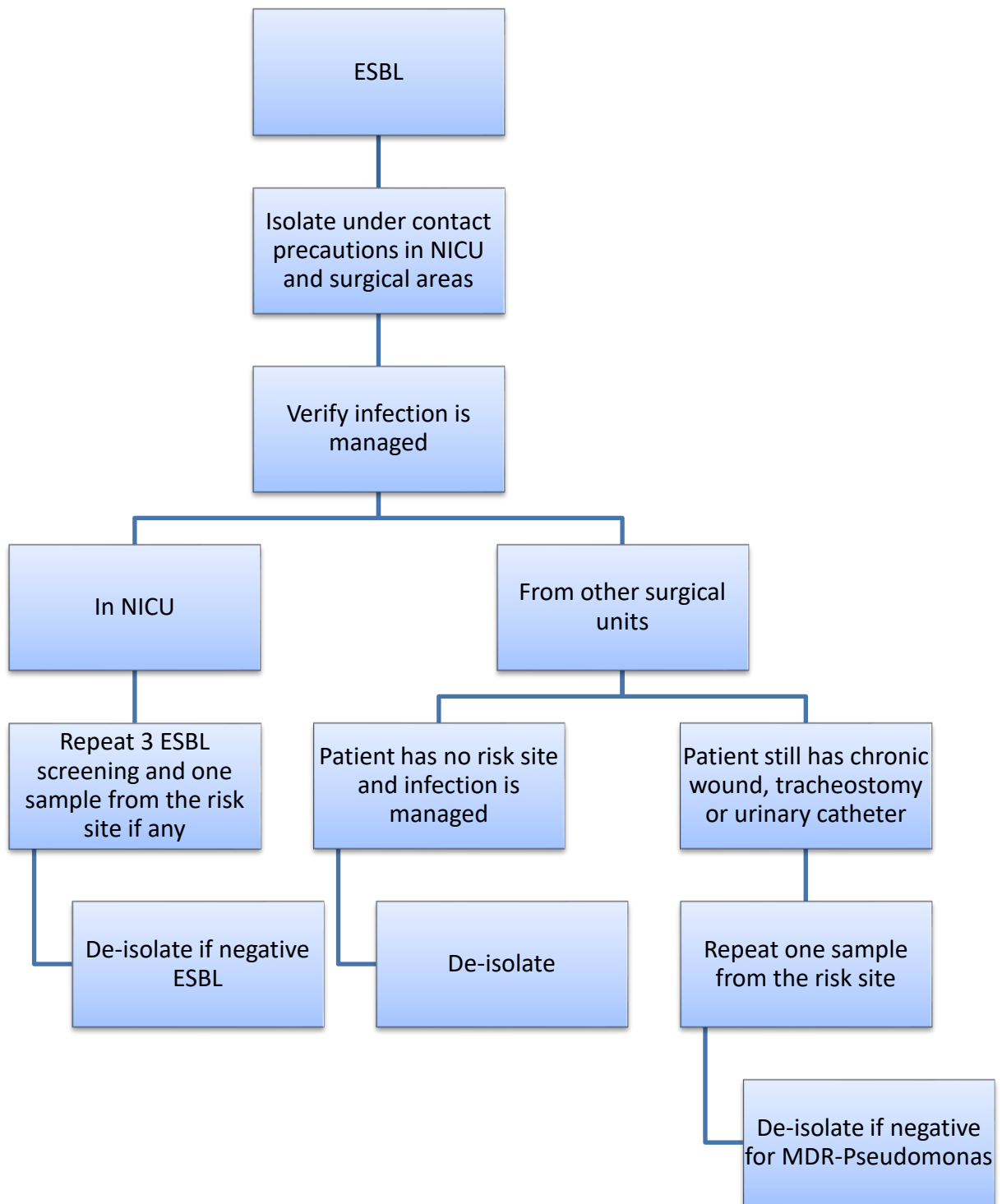


Figure 3 : De-isolation of patients with ESBL organisms. Note that the same algorithm is applicable for Serratia in the NICU.

#### **7.4 Screening patient for the purpose of shifting to other hospital:**

Patients can be screened before shifting if requested by the receiving hospital. In case the patient is on antibiotics inform the receiving hospital that screening was done while patient was on antibiotics.

#### **7.5 Contact screening:**

- **MDRO Contact**
- Screening of contacts is only required if the index patient is MDRO positive and the patient has not been isolated on admission, or during an outbreak situation.
- A patient who has shared the same bed area for more than 24 hours with MDRO colonised or infected patient where contact precaution were not in place
- Screening of patients in the same setting is not normally required if the case was identified on admission and the patient isolated immediately
- Screening of household contacts and healthcare staff is not required. There is no compelling evidence to suggest that screening the household contacts or healthcare workers to check for colonisation will provide additional benefit in controlling spread in the healthcare setting

#### **7.6 Screening during an outbreak:**

As advised by outbreak management team & IP&C department depending on the micro-organism involved.

## **7. Specimen Collection Procedure:**

### **8.1 General instructions:**

- Please note that privacy and comfort should be assured before commencing sample collection.
- Explain to the patient the procedure before initiating the screening
- Before collection hand hygiene should be performed and then disposable gloves should be donned.
  
- Pre-moisten all swabs with sterile normal saline prior to taking a specimen.
- Swab wounds, skin lesions, incisions, ulcers or central line exit site if present with separate swabs.
- All swabs collected should correctly labelled with patient sticker or hospital number, screened site and MDRO Code (MRSA, CRE, MDRAB, VRE, etc.) for screening
- Each MDRO of interest should have a separate swab collected.
- Please note that of all body sites being swabbed the rectum should always be the last in the swabbing sequence.

### **8.2 Collection Procedure**

#### **8.2.1 Nasal swab**

- Open and remove sterile tipped swab applicator from transport casing and pre-moisten it.
- Taking care to avoid other contact with swab, insert the swab approximately 1-2 cm (approximately  $\frac{3}{4}$  inches) into the first nostril next to the nasal septum.
- Rotate the swab against the anterior nasal mucosa for 3 – 5 seconds.
- Using the same swab, repeat for the other nostril.
- Carefully place used swab back into transport tube and secure.

### **8.2.2 Axilla & groin swabs:**

- Clean the site with normal saline to remove debris before swabbing.
- Open and remove sterile tipped swab applicator from transport casing and pre-moisten it.
- Use the same swab for identical sites: one swab for both axilla and one swab for both inguinal areas.

### **8.2.3 Rectal swab**

- Ask the patient to loosen their clothing
- Open and remove sterile tipped swab applicator from transport casing and pre-moisten it.
- Taking care to avoid other contact with swab, rotate the swab against the external rectal orifice. If visible stool is not obtained on the swab, insert it a few millimetres into the rectum until visible stool is obtained.
- Carefully place used swab back into transport tube and secure.

### **8.2.4 Other samples:**

For sample collection from other risk sites such as wounds, ET secretion and urine follow microbiology laboratory guidance.

## **8. Acting on Results of Screening / Isolation of Patients:**

Any patient screened for MDROs and results became positive follow isolations policy for further actions.

## 9. References:

| <b>Title of book/ journal/ articles/ Website</b>                                                     | <b>Author</b>                                     | <b>Year of publication</b> | <b>Page</b> |
|------------------------------------------------------------------------------------------------------|---------------------------------------------------|----------------------------|-------------|
| Screening, Testing and Surveillance for Antibiotic-Resistant Organisms (AROs) in healthcare settings | Provincial Infectious Diseases Advisory Committee | 2.2013                     | 7, 40-47    |
| Methicillin Resistant Staphylococcus Aureus (MRSA) Protocol                                          | King's College Hospital NHS trust, UK             | 9.2011                     | 4-5,7       |
| Protocol for CRA MRSA Screening National Rollout in Scotland                                         | Health Protection Scotland                        | 31.1.2013                  | 3-4, 11-15  |
| GCC infection prevention & control Manual                                                            | GCC Center of IC                                  | 2013                       | 108-109     |



**10.Review / Revision History:**

| Revision History |                                                                                                                          |                                        |                 |
|------------------|--------------------------------------------------------------------------------------------------------------------------|----------------------------------------|-----------------|
| Revision         | Description of change                                                                                                    | Author                                 | Revision Date   |
| 1                | Initial Release                                                                                                          | Amina AL Jardani                       | 16/11/2010      |
| 2                | Reviewed                                                                                                                 | Zaina AL Maskari                       | 16/11/2016      |
| 3                | Reviewed                                                                                                                 | Ahlam Al Baloushi<br>Khadija AlZadjali | 20/04/2022      |
| 4                | Reviewed                                                                                                                 | Maryam Al Balushi                      | 12/03/2023      |
| 5                | Changed the criteria for MDRO screening in the NICU.<br><br>Added algorithm for de-isolation of ESBL and MDR pseudomonas | Omaira Al Omairi                       | 4/11/2024       |
| Written by       |                                                                                                                          | Checked by                             | Authorized by   |
| Zaina AL Maskari |                                                                                                                          | Dr. Omaira Al Omairi                   | Dr. Amal Al Tai |